



Objectives:

- Need of Line Coding
- Introduction of Line Coding
- Properties of Line Coding
- Types of Line Coding
- Advantages and Disadvantages
- Power Spectral Density
- PSD of Line Coding
- Comparison of Line Coding



Need Of Line Coding:

- Various Techniques
- Other Way: From Computers
- Information: Inherently discrete in nature
- Transmitted over band-limited channel: Signal gets Dispersed
- Causes: Overlap and Distortion
- Distortion: Intersymbol Interference(ISI)

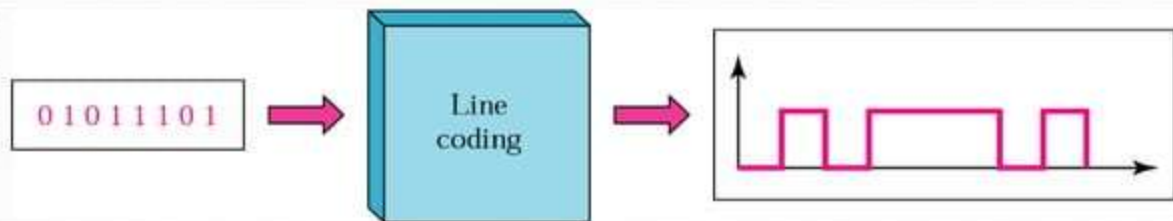


To avoid all these problems we are going for

Line Coding

Introduction:

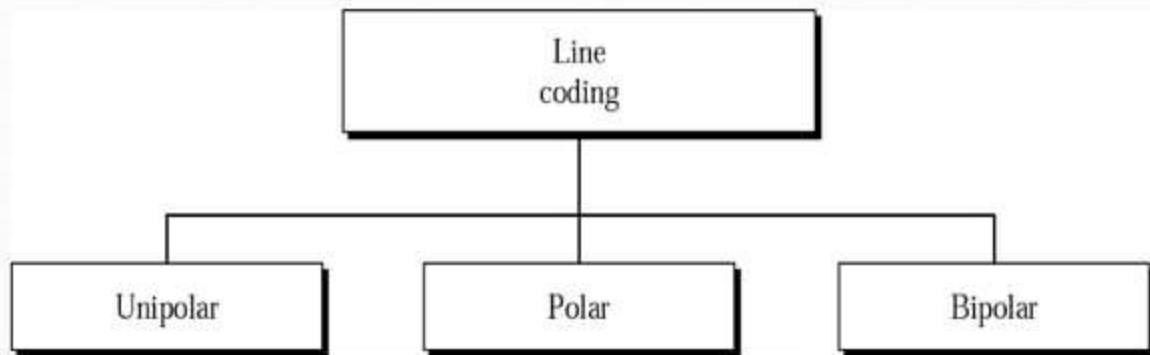
- Binary Data: Pulses
- Line Coding: A pair of pulses to represent symbols 1 and 0



Properties of Line Coding:

- Transmission Bandwidth: as small as possible
- Power Efficiency: As small as possible for given BW and probability of error
- Error Detection and Correction capability: Ex: Bipolar
- Favorable power spectral density: $dc=0$
- Adequate timing content: Extract timing from pulses
- Transparency: Prevent long strings of 0s or 1s

Types of Line Coding:





Unipolar Signaling:

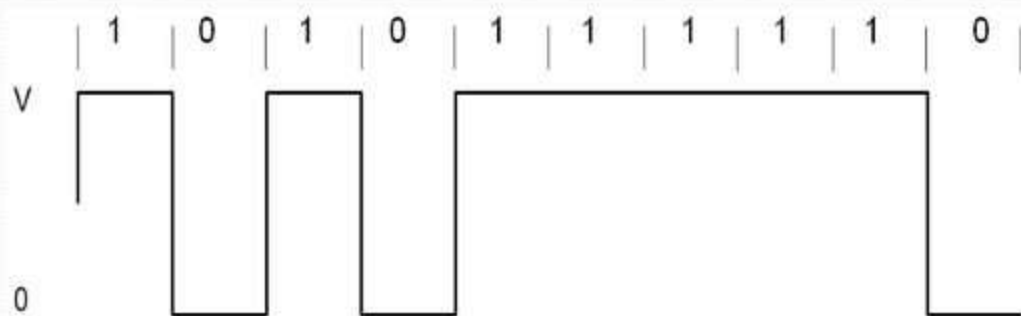
- On-Off keying ie OOK
- Pulse 0: Absence of pulse
- Pulse 1 : Presence of pulse

There are two common variations of unipolar signalling:

1. Non-Return to Zero (NRZ)
2. Return to Zero (RZ)

Unipolar Non-Return to Zero (NRZ):

- Duration of the MARK pulse ($\mathcal{T}$) is equal to the duration (T_o) of the symbol slot.



Advantages:

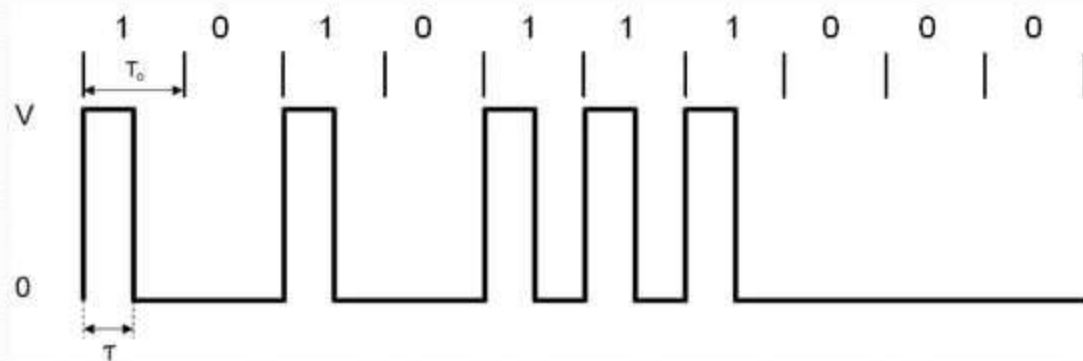
- Simplicity in implementation
- Doesn't require a lot of bandwidth for transmission.

Disadvantages:

- Presence of DC level (indicated by spectral line at 0 Hz).
- Contains low frequency components. Causes "Signal Droop"
- Does not have any error correction capability.
- Does not possess any clocking component for ease of synchronisation.
- Is not Transparent. Long string of zeros causes loss of synchronisation.

Unipolar Return to Zero (RZ):

- MARK pulse ($\mathcal{T}$) is **less** than the duration (T_o) of the symbol slot.
- Fills only the first half of the time slot, returning to zero for the second half.





Advantages:

- Simplicity in implementation.
- Presence of a spectral line at symbol rate which can be used as symbol timing clock signal.

Disadvantages:

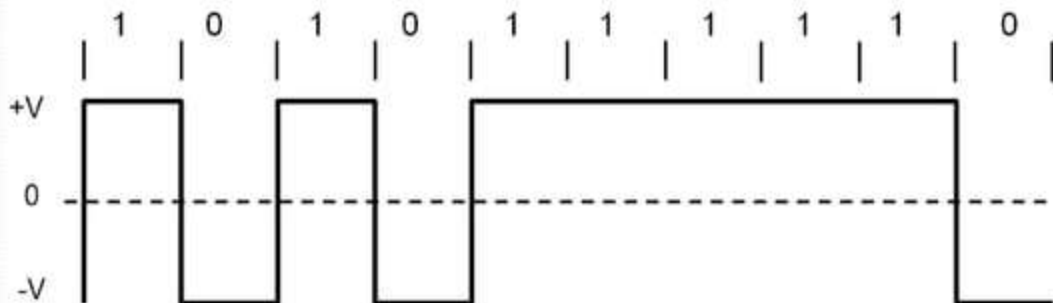
- Presence of DC level (indicated by spectral line at 0 Hz).
- Continuous part is non-zero at 0 Hz. Causes “Signal Droop”.
- Does not have any error correction capability.
- Occupies twice as much bandwidth as Unipolar NRZ.
- Is not Transparent

Polar Signalling:

- Polar RZ
- Polar NRZ

Polar NRZ:

- A binary 1 is represented by a pulse $g_1(t)$
- A binary 0 by the opposite (or antipodal) pulse $g_0(t) = -g_1(t)$.





Advantages:

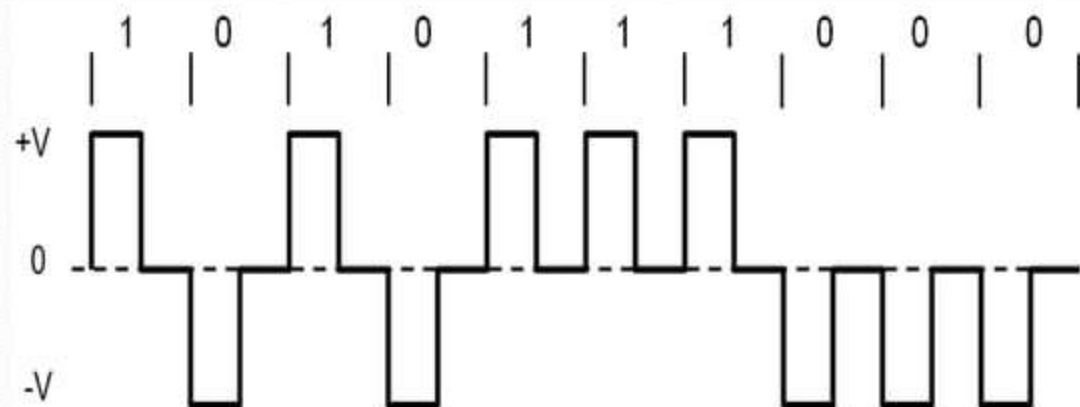
- Simplicity in implementation.
- No DC component.

Disadvantages:

- Continuous part is non-zero at 0 Hz. Causes “Signal Droop”.
- Does not have any error correction capability.
- Does not possess any clocking component for ease of synchronisation.
- Is not transparent.

Polar RZ:

- A binary 1: A pulse $g_1(t)$
- A binary 0: The opposite (or antipodal) pulse $g_0(t) = -g_1(t)$.
- Fills only the first half of the time slot, returning to zero for the second half.





Advantages:

- Simplicity in implementation.
- No DC component.

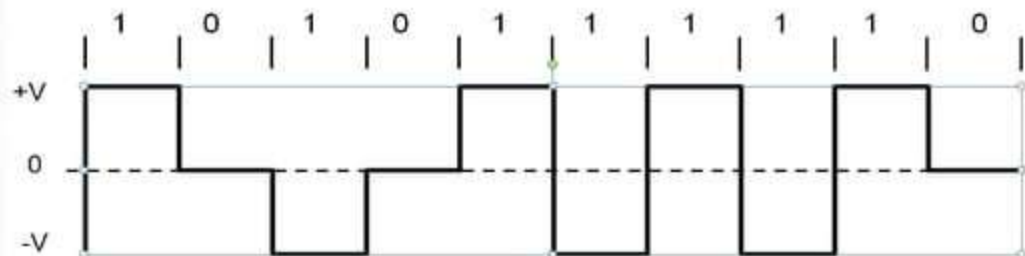
Disadvantages:

- Continuous part is non-zero at 0 Hz. Causes “Signal Droop”.
- Does not have any error correction capability.
- Occupies twice as much bandwidth as Polar NRZ.

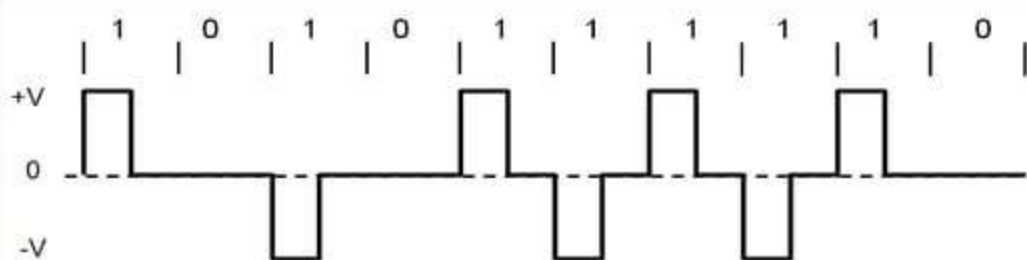
Bipolar Signalling:

- Alternate mark inversion (AMI)
- Uses three voltage levels ($+V$, 0 , $-V$)
- 0 : Absence of a pulse
- 1 : Alternating voltage levels of $+V$ and $-V$

Bipolar NRZ:



Bipolar RZ:





Advantages:

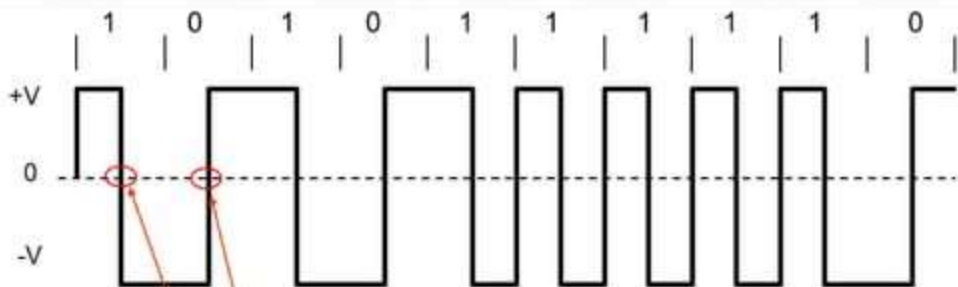
- No DC component.
- Occupies less bandwidth than unipolar and polar NRZ schemes.
- Does not suffer from signal droop (suitable for transmission over AC coupled lines).
- Possesses single error detection capability.

Disadvantages:

- Does not possess any clocking component for ease of synchronisation.
- Is not Transparent.

Manchester Signalling:

- The duration of the bit is divided into two halves
- A 'One' is +ve in 1st half and -ve in 2nd half.
- A 'Zero' is -ve in 1st half and +ve in 2nd half.



Note: There is always a transition at the centre of bit duration.



Advantages:

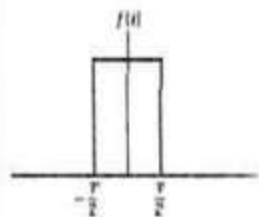
- No DC component.
- Does not suffer from signal droop (suitable for transmission over AC coupled lines).
- Easy to synchronise.
- Is Transparent.

Disadvantages:

- Because of the greater number of transitions it occupies a significantly large bandwidth.
- Does not have error detection capability.

Power Spectral Density:

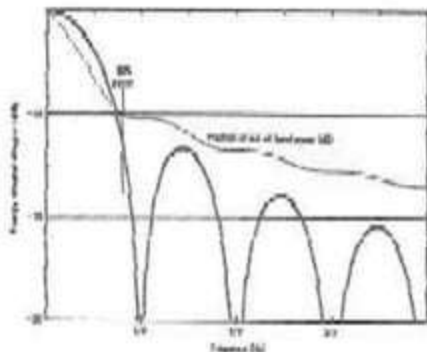
- The function which gives distribution of power of a signal at various frequencies in frequency domain.
- PSD is the Fourier Transform of autocorrelation
- Rectangular pulse and its spectrum



$$f(t) = \begin{cases} 1 & |t| \leq \frac{T}{2} \\ 0 & \text{otherwise} \end{cases}$$

$$F(w) = (T) \frac{\sin(wT/2)}{wT/2}$$

where w = radian frequency $2\pi f$,
 T = duration of a signal interval.



PSD Derivation:

- We now need to derive the time autocorrelation of a power signal $x(t)$

$$R_x(\tau) = \lim_{T_p \rightarrow \infty} \frac{1}{T_p} \int_{-T_p/2}^{T_p/2} x(t)x(t+\tau)dt$$

- Since $x(t)$ consists of impulses, $R_x(\tau)$ is found by

$$R_x(\tau) = \frac{1}{T} \sum_{n=-\infty}^{\infty} R_n \delta(\tau - nT)$$

where $R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n}$

- Recognizing $R_n = R_{-n}$ for real signals, we have

$$S_x(w) = \frac{1}{T} \left(R_0 + 2 \sum_{n=1}^{\infty} R_n \cos nwT \right)$$

- Since the pulse filter has the spectrum of $F(w) \leftrightarrow f(t)$, we have

$$\begin{aligned} S_y(w) &= |F(w)|^2 S_x(w) \\ &= |F(w)|^2 \left(\sum_{n=-\infty}^{\infty} R_n e^{-jnwT_b} \right) \\ &= \frac{|F(w)|^2}{T} \left(R_0 + 2 \sum_{n=1}^{\infty} R_n \cos nwT \right) \end{aligned}$$

- Now, we can use this to find the PSD of various line codes.

PSD of Polar Signalling:

- In polar signalling,
Binary “1” is transmitted by a pulse $f(t)$
Binary “0” is transmitted by a pulse $-f(t)$
- In this case, a_k is equally likely to be 1 or -1 and a_k^2 is always 1.

$$R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k^2 = \lim_{N \rightarrow \infty} \frac{1}{N} (N) = 1$$

Where, There are N pulses and $a_k^2=1$ for each one.

The summation on the right-hand side of the above equation is N .

- Moreover, both a_k and a_{k+1} are either 1 or -1. So, $a_k a_{k+1}$ is either 1 or -1.

They are equally likely to be 1 or -1 on the average, out of N terms the product $a_k a_{k+1}$ is equal to 1 for $N/2$ terms and is equal to -1 for the remaining $N/2$ terms.

$$R_1 = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{2}(1) + \frac{N}{2}(-1) \right] = 0 \quad S_y(w) = \frac{|F(w)|^2}{T} R_0 = \frac{|F(w)|^2}{T}$$

$$R_n = 0 \quad n \geq 1 \quad S_y(w) = \frac{T}{2} \text{sinc}^2 \left(\frac{wT}{2} \right)$$

PSD of Bipolar Signalling:

- To calculate the PSD, we have

$$R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k a_{k+n} \quad R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k^2$$

- On the average, half of the a_k 's are 0, and the remaining half are either 1 or -1, with $a_k^2=1$. Therefore,

$$R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{2} (\pm 1)^2 + \frac{N}{2} (0)^2 \right] = \frac{1}{2}$$

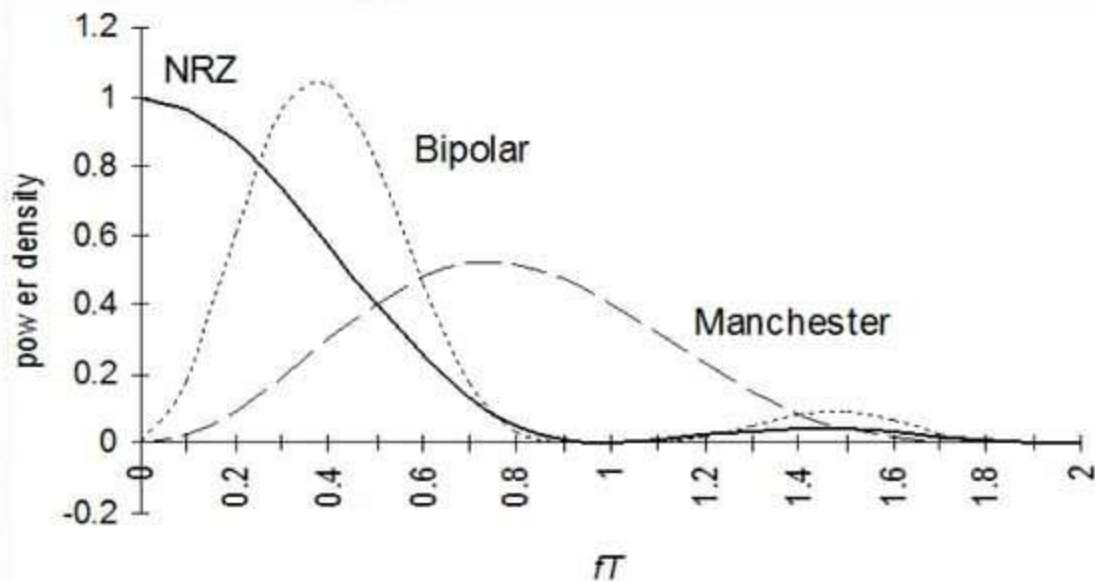
•To compute R_1 , we consider the pulse strength product $a_k a_{k+1}$.

-Four possible equally likely sequences of two bits: 11, 10, 01, 00.

-Since bit 0 encoded by no pulse ($a_k=0$), the product $a_k a_{k+1}=0$ for the last three of these sequences. This means that, on the average, $3N/4$ combinations have $a_k a_{k+1}=0$ and only $N/4$ combinations have non zero $a_k a_{k+1}$. Because of the bipolar rule, the bit sequence 11 can only be encoded by two consecutive pulse of opposite polarities. This means the product $a_k a_{k+1} = -1$ for the $N/4$ combinations.

$$R_1 = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{4}(-1) + \frac{N}{4}(0) \right] = -\frac{1}{4}$$

PSD of Lines Codes:



Comparison of Line Codes:

Sr. No.	Parameters	Polar RZ	Polar NRZ	AMI	Manchester
1	Transmission of DC component	YES	YES	NO	NO
2	Signaling Rate	1/Tb	1/Tb	1/Tb	1/Tb
3	Noise Immunity	LOW	LOW	HIGH	HIGH
4	Synchronizing Capability	Poor	Poor	Very Good	Very Good
5	Bandwidth Required	1/Tb	1/2Tb	1/2Tb	1/Tb
6	Crosstalk	HIGH	HIGH	LOW	LOW



THANK YOU